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**Ex 33: Find the Number of Vowels in Prolog:**

**PROLOG CODE:**

```
is_vowel(a). is_vowel(e). is_vowel(i). is_vowel(o). is_vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-
   is_vowel(H),
   count_vowels(T, RestCount),
   Count is RestCount + 1.

count_vowels([H|T], Count) :-
   \+ is_vowel(H),
   count_vowels(T, Count).
```

**OUTPUT:**

```
39 ?- [user].
|: is_vowel(a). is_vowel(e). is_vowel(i). is_vowel(o). is_vowel(u).
|:
|:
|:
|: count_vowels([], 0).
|:
|: count_vowels([H|T], Count) :-
|:
|:     is_vowel(H),
|:
|:     count_vowels(T, RestCount),
|:
|:     Count is RestCount + 1.
|:
|: count_vowels([H|T], Count) :-
|:
|:     \+ is_vowel(H),
|:
|:     count_vowels(T, Count).
|: ^D% user://14 compiled 0.00 sec, 8 clauses
true.

40 ?- count_vowels([a, i, c, s, e], Count).
Count = 3
```